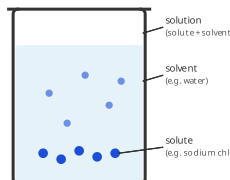


QUANTITATIVE CHEMISTRY 2 – CONCENTRATION OF SOLUTIONS

Progress: ○ ○ ○ ○

WHAT IS A SOLUTION?



The **solute** is the substance being dissolved.
The **solvent** is the liquid that does the dissolving.

THE FORMULA YOU NEED

$$\text{concentration (g/dm}^3\text{)} = \frac{\text{mass of solute (g)}}{\text{volume of solution (dm}^3\text{)}}$$

CONVERSION REMINDER

$$1000 \text{ cm}^3 = 1 \text{ dm}^3$$

WORKED EXAMPLE (FOLLOW THIS!)

12 g of sodium chloride is dissolved in 0.75 dm³ of water.

Find the concentration.

Step 1: Write the formula:

concentration = mass ÷ volume

Step 2: Put in the numbers:

= 12 g ÷ 0.75 dm³

Step 3: Answer = 16 g/dm³

1 CONVERT THE UNITS

Convert 250 cm³ to dm³.

(Look at the conversion reminder above.)

- ☐ Remember: to go from cm³ to dm³, you divide by 1000.
- ☐ 250 ÷ 1000 = ?

Answer: _____ dm³

2 FIND THE CONCENTRATION

A student dissolves 5.0 g of copper sulfate in 0.50 dm³ of water.

Work out the concentration in g/dm³.

- ☐ Write the formula: concentration = mass ÷ volume
- ☐ Put in the numbers: 5.0 ÷ 0.50
- ☐ Calculate and write the unit (g/dm³)

Answer: _____ g/dm³

3 FIND THE MASS

A solution has a concentration of 24 g/dm³. The volume is 0.40 dm³.

Work out the mass of solute.

- ☐ Rearrange: mass = concentration × volume
- ☐ Put in the numbers: 24 × 0.40
- ☐ Calculate and write the unit (g)

Answer: _____ g

4 WHICH IS STRONGER?

Two solutions are made.

Solution	Mass of solute (g)	Volume of solution (dm ³)
A	8.0	0.40
B	12.0	0.75

Which solution is **more concentrated**? Circle your answer.

A B

Show your working.

- ☐ Calculate concentration of A: 8.0 ÷ 0.40 = ?
- ☐ Calculate concentration of B: 12.0 ÷ 0.75 = ?
- ☐ Which number is bigger? Circle that one.